

Bootstrap: name based on the story of Baron Munchausen pulling himself and his horse out of a swamp by his bootstrap (or pigtail depending on the version).



- If we want to avoid this normal approximation, we can use the following algorithm:
  - Sample  $n$  observations  $x_1^*, \dots, x_n^*$  with replacement from the observed values  $x_1, \dots, x_n$  (this is called a bootstrap sample).
  - Compute the average in the bootstrap sample, i.e.

$$\bar{x}^* = \frac{1}{n} \sum_{i=1}^n x_i^*.$$

- Repeat these steps  $B$  times to obtain  $\bar{x}_1^*, \dots, \bar{x}_B^*$ .
  - Use the  $\alpha/2$  and  $1 - \alpha/2$  quantiles from the sample of averages  $\bar{x}_1^*, \dots, \bar{x}_B^*$  as lower and upper bound of the confidence interval.
- The bootstrap percentile confidence interval can be used for any estimator  $\hat{\vartheta}$  for any parameter  $\vartheta$  by replacing the arithmetic mean  $\bar{X}$  with this estimator. This approach usually works well if the bootstrap distribution is symmetric around the observed  $\hat{\vartheta}$ .

- The underlying idea is as follows: If we could repeatedly draw random samples from the true population model (instead of drawing with replacement from the observed sample), we could directly estimate the distribution of  $\bar{X}$ , or more generally  $\hat{\vartheta}$ .
- Since the true population distribution is not available, we approximate it by the empirical distribution, where each data point is assigned probability  $1/n$ .

## Empirical Cumulative Distribution Function:

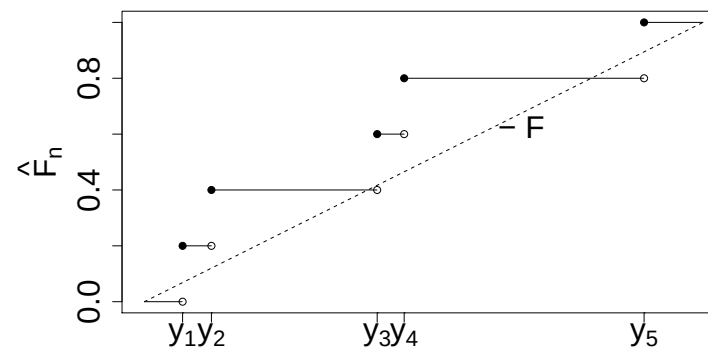
- The empirical cumulative distribution function (ecdf)

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}(x_i \leq x)$$

is a good, nonparametric estimate for the true cumulative distribution function

$$F(x) = P(X \leq x)$$

regardless of the type and structure of  $F(x)$ .

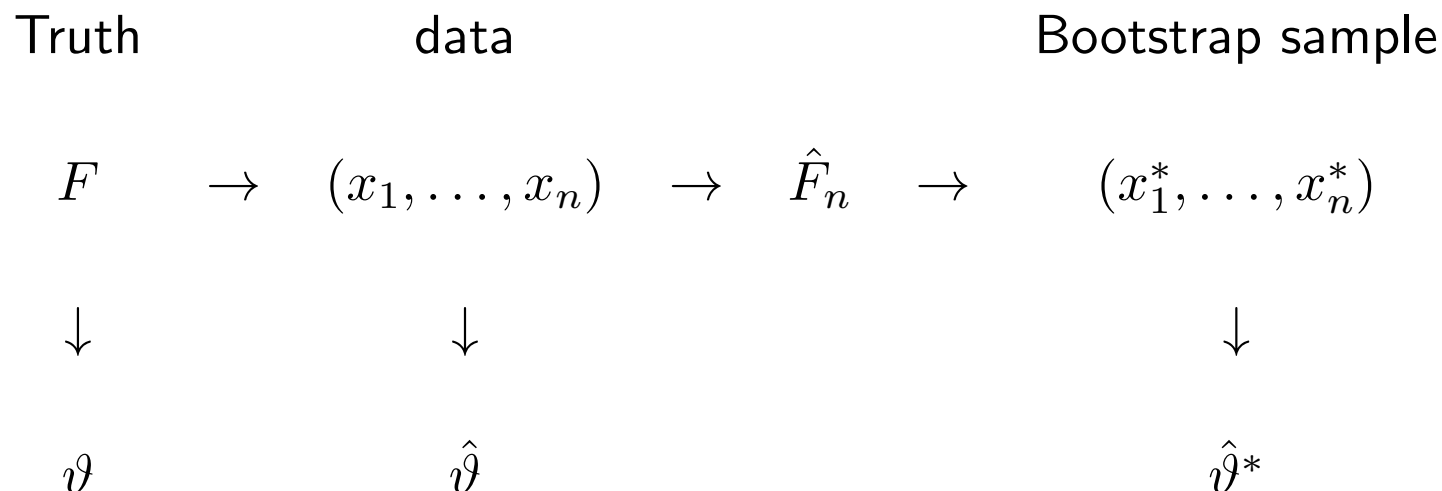


- In particular, we have

$$\hat{F}_n(x) \rightarrow F(x) \text{ as } n \rightarrow \infty \text{ for all } x \text{ (almost surely)}$$

i.e.  $\hat{F}_n(x)$  converges to  $F(x)$  for growing sample size.

- Hence, we can safely replace the true population model with the empirical distribution (if  $n$  is not too small).



Two approximations:

- $F$  approximated by  $\hat{F}_n$ . **Statistical error** due to variability in the data and finite  $n$ , independent of  $B$ .
- Theoretical quantities such as  $E^*(\hat{\vartheta}^*)$  approximated by estimators such as  $\frac{1}{B} \sum_{b=1}^B \hat{\vartheta}^{*b}$ . **Simulation error** due to a finite  $B$ , reduces when increasing  $B$ ! Is zero if for small  $n$  all possible subsamples can be taken.

## Remarks:

- Drawing a random sample of size  $n$  from  $\hat{F}_n(x)$  is equivalent to sampling with replacement from the observed sample since  $\hat{F}_n(x)$  assigns probability  $1/n$  to each observed data point.
- Since we draw with replacement, a bootstrap sample  $x_1^*, \dots, x_n^*$  will in general differ from  $x_1, \dots, x_n$  and will contain duplicates of some of the observations (and no copies of others).
- The bootstrap does not require asymptotic assumptions on the distribution of the estimator.
- It assumes that  $\hat{F}_n(x)$  is a good approximation of  $F(x)$ , which again relies on an asymptotic argument, but due to different orders of convergence bootstrap based quantities can have an asymptotic advantage over those using asymptotic normality.
- The bootstrap confidence interval is particularly useful for estimators where no standard results for the construction of confidence intervals are available.

**Example:** Confidence interval for the correlation coefficient.

- Let  $(X_1, Y_1), \dots, (X_n, Y_n)$  be i.i.d. replicates from a bivariate random vector  $(X, Y)$ .
- We are interested in a confidence interval for the correlation  $\rho$ , which we estimate with the correlation coefficient

$$\hat{\rho} = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}.$$

- Generate  $B$  bootstrap samples  $(x_1^{*b}, y_1^{*b}), \dots, (x_n^{*b}, y_n^{*b})$ ,  $b = 1, \dots, B$  by sampling with replacement from the observed data pairs.
- Determine  $\hat{\rho}^{*b}$  for each of the bootstrap samples and use the  $\alpha/2$  and  $1 - \alpha/2$  quantiles for constructing the confidence interval.



## Bootstrap Estimates for Bias and Variance:

- The bootstrap can also be used to derive other properties of an estimator  $\hat{\vartheta}$ , e.g. estimates for its bias and variance.
- Assume that  $X_1, \dots, X_n$  are i.i.d. and we are interested in a parameter  $\vartheta$ .
- Based on bootstrap samples  $x_1^{*b}, \dots, x_n^{*b}$  with estimates  $\hat{\vartheta}^{*b}$ ,  $b = 1, \dots, B$ , we can estimate
  - the expected value of  $\hat{\vartheta}$  as

$$\hat{E}_B(\hat{\vartheta}) = \frac{1}{B} \sum_{b=1}^B \hat{\vartheta}^{*b}.$$

- the bias of  $\hat{\vartheta}$  as

$$\widehat{\text{Bias}}_B(\hat{\vartheta}) = \hat{E}_B(\hat{\vartheta}) - \hat{\vartheta}.$$

– the variance of  $\hat{\vartheta}$  as

$$\widehat{\text{Var}}_B(\hat{\vartheta}) = \frac{1}{B-1} \sum_{b=1}^B \left( \hat{\vartheta}^{*b} - \hat{E}_B(\hat{\vartheta}) \right)^2.$$

## Alternative Bootstrap Confidence Intervals:

- Asymptotic confidence interval: Consider

$$Z = \frac{\hat{\vartheta} - \vartheta}{\sqrt{\widehat{\text{Var}}_B(\hat{\vartheta})}} \stackrel{a}{\sim} N(0, 1)$$

as an asymptotic pivot to obtain the  $1 - \alpha$  confidence interval

$$\hat{\vartheta} \pm z_{1-\frac{\alpha}{2}} \sqrt{\widehat{\text{Var}}_B(\hat{\vartheta})}.$$

Idea: Instead of the asymptotic variance of the ML estimator, we use the bootstrap estimate for the variance.

- Bootstrap t-interval: To avoid the assumption of asymptotic normality for  $Z$ , determine estimated quantiles  $\hat{z}_\alpha$  also from the bootstrap.

- Generate

$$z_b^* = \frac{\hat{\vartheta}^{*b} - \hat{\vartheta}}{\sqrt{\text{Var}(\hat{\vartheta}^{*b})}}$$

for each of the  $B$  bootstrap samples.

- Estimate the quantiles  $\hat{z}_{\frac{\alpha}{2}}$  and  $\hat{z}_{1-\frac{\alpha}{2}}$  as empirical quantiles from  $z_1^*, \dots, z_B^*$ .
- The bootstrap t-interval is then given by

$$\left[ \hat{\vartheta} - \hat{z}_{1-\frac{\alpha}{2}} \sqrt{\widehat{\text{Var}}_B(\hat{\vartheta})}, \hat{\vartheta} - \hat{z}_{\frac{\alpha}{2}} \sqrt{\widehat{\text{Var}}_B(\hat{\vartheta})} \right].$$

- Analogy to the (symmetric) t-confidence interval:

$$\left[ \hat{\vartheta} - t_{1-\frac{\alpha}{2}} \sqrt{\widehat{\text{Var}}_B(\hat{\vartheta})}, \hat{\vartheta} + t_{1-\frac{\alpha}{2}} \sqrt{\widehat{\text{Var}}_B(\hat{\vartheta})} \right]$$

with  $t_{1-\frac{\alpha}{2}} = -t_{\frac{\alpha}{2}}$  the  $1 - \frac{\alpha}{2}$  quantile of the corresponding t-distribution.

– Intuition:

$$\begin{aligned}
 1 - \alpha &= P\left(\hat{z}_{\frac{\alpha}{2}} \leq \frac{\hat{\vartheta}^{*b} - \hat{\vartheta}}{\sqrt{\text{Var}(\hat{\vartheta}^{*b})}} \leq \hat{z}_{1-\frac{\alpha}{2}}\right) \\
 &= P\left(\hat{\vartheta}^{*b} - \hat{z}_{1-\frac{\alpha}{2}}\sqrt{\text{Var}(\hat{\vartheta}^{*b})} \leq \hat{\vartheta} \leq \hat{\vartheta}^{*b} - \hat{z}_{\frac{\alpha}{2}}\sqrt{\text{Var}(\hat{\vartheta}^{*b})}\right)
 \end{aligned}$$

in the bootstrap world corresponds to

$$1 - \alpha \approx P\left(\hat{\vartheta} - \hat{z}_{1-\frac{\alpha}{2}}\sqrt{\widehat{\text{Var}}_B(\hat{\vartheta})} \leq \vartheta \leq \hat{\vartheta} - \hat{z}_{\frac{\alpha}{2}}\sqrt{\widehat{\text{Var}}_B(\hat{\vartheta})}\right)$$

for the truth using the bootstrap **plug-in principle** ( $\hat{\vartheta}$  for  $\vartheta$ ,  $\hat{\vartheta}^{*b}$  for  $\hat{\vartheta}$  etc.).

- This approach has certain optimality properties, as the bootstrapped quantity  $Z$  is a pivot, unlike in the percentile based approach, and as it does not use normality, unlike the asymptotic version.

- The generation of  $z_b^*$  requires the variance  $\text{Var}(\hat{\vartheta}^{*b})$  of the estimator in the  $b$ -th bootstrap sample. If this variance can not be calculated analytically, perform a second bootstrap within the  $b$ -th bootstrap to estimate the variance, i.e. a *nested bootstrap*.

### Number of Bootstrap Replications:

- For estimating variance and bias,  $B = 200$  bootstrap samples are typically sufficient.
- For estimating quantiles (as required for the bootstrap percentile interval or the bootstrap t-interval),  $B = 2000$  is usually recommended.

### Further reading:

- Davison and Hinkley (1997). Bootstrap Methods and their Applications. Cambridge.
- Efron and Tibshirani (1993). An Introduction to the Bootstrap. Springer.